Relative Grading System — Complete Report

Introduction

Relative grading is a modern grading system used in many universities, especially engineering universities.

In University of Engineering and Technology Peshawar and other universities, students are not graded only on fixed percentages.
Instead, grades are assigned by comparing the performance of students in the whole class.

This means:

- Class average is important
- Student performance is compared with other students
- Grades may change depending on class performance

For example:

- If the paper is difficult and most students get low marks, students with better performance can still receive high grades.

Absolute Grading vs Relative Grading

1. Absolute Grading

In absolute grading, grades are fixed.

Marks	Grade
90 - 100	A
80 - 89	B
70 - 79	C

In this system, class performance does not matter.

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2. Relative Grading

In relative grading:

- Class average matters
- Students are compared with each other
- Grades are assigned according to average performance

Example:

If:

- Class average = 45
- Student marks = 70

The student may receive grade A because the marks are much higher than the average.

Objective of This Project

The objectives of this project are:

✅ Take marks input from students
✅ Calculate class average
✅ Find highest and lowest marks
✅ Apply relative grading
✅ Display final result

Programming Concepts Used

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1. Arrays

What is an Array?

An array is a variable that stores multiple values of the same type.

Example:

```
int marks[5];
```

This array stores marks of 5 students.

Array Representation

Index	Value
0	85
1	70
2	60
3	45
4	30

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2. Functions

What is a Function?

Functions divide a large program into smaller parts.

Benefits:

- Makes code reusable
- Makes program easier
- Reduces errors

Functions Used in This Program

Function	Purpose
`inputMarks()`	Takes marks input
`calculateAverage()`	Calculates average
`findMaximum()`	Finds highest marks
`findMinimum()`	Finds lowest marks
`assignGrade()`	Assigns relative grade
`displayResult()`	Displays final result

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3. Loops

Loops are used to repeat tasks.

Examples:

- Taking marks input
- Displaying grades
- Calculating total marks

For Loop Structure

for(initialization; condition; increment)\texttt{for(initialization; condition; increment)}for(initialization; condition; increment)

Example

```
for(int i = 0; i < n; i++)
```

Working of This Loop

Step 1

```
int i = 0;
```

The loop starts from 0.

Step 2

```
i < n;
```

The loop continues until the condition becomes false.

Step 3

```
i++;
```

The value of i increases after every iteration.

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4. Conditional Statements

Conditional statements are used for decision making.

Example:

```
if(marks >= average + 15)
```

If marks are 15 greater than average:

- Grade A is assigned.

Relative Grading Logic

Condition	Grade
Average + 15 or more	A
Average + 5	B
Near average	C
Below average	D
Very low marks	F

Program Flow

Step-by-Step Working

Step 1

The user enters the number of students.

Example:

```
Enter number of students: 5
```

Step 2

The program takes marks of all students.

Step 3

The program calculates average.

Formula:

Average=Sum of all marksTotal students\text{Average} = \frac{\text{Sum of all marks}}{\text{Total students}}Average=Total studentsSum of all marks

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Step 4

The program finds:

- Maximum marks
- Minimum marks

Step 5

Relative grading is applied.

Step 6

Final result is displayed.

COMPLETE C++ PROGRAM WITH ROMAN URDU COMMENTS

```
#include <iostream>
using namespace std;

// ====================================================
```

```
// Function Prototypes
// ======================================================

// Marks input lene ka function
void inputMarks(int marks[], int n);

// Average calculate karne ka function
float calculateAverage(int marks[], int n);

// Maximum marks find karne ka function
int findMaximum(int marks[], int n);

// Minimum marks find karne ka function
int findMinimum(int marks[], int n);

// Relative grading assign karne ka function
char assignGrade(int marks, float average);

// Final result display karne ka function
void displayResult(int marks[], int n, float average);

// ======================================================
// Main Function
// ======================================================

int main()
{
    // Roman Urdu:
    // n variable students ki total quantity store karega

    int n;

    cout << "Enter number of students: ";
    cin >> n;

    // Roman Urdu:
    // Array students ke marks store karega

    int marks[n];

    // Marks input function call
    inputMarks(marks, n);

    // Average calculate karna
    float average = calculateAverage(marks, n);

    // Maximum marks find karna
    int maximum = findMaximum(marks, n);
```

```
    // Minimum marks find karna
    int minimum = findMinimum(marks, n);

    // Final result display karna
    displayResult(marks, n, average);

    cout << "\n============================\n";

    cout << "Class Average = " << average << endl;
    cout << "Maximum Marks = " << maximum << endl;
    cout << "Minimum Marks = " << minimum << endl;

    cout << "============================\n";

    return 0;
}

// =====================================================
// Function: inputMarks()
// =====================================================

void inputMarks(int marks[], int n)
{
    // Roman Urdu:
    // Ye loop har student ke marks input lene ke liye use ho raha hai

    for(int i = 0; i < n; i++)
    {
        cout << "Enter marks of student "
             << i + 1 << ": ";

        cin >> marks[i];
    }
}

// =====================================================
// Function: calculateAverage()
// =====================================================

float calculateAverage(int marks[], int n)
{
    // Roman Urdu:
    // sum variable total marks store karega

    int sum = 0;

    // Roman Urdu:
    // Sare marks add karne ke liye loop use kiya
```

```
for(int i = 0; i < n; i++)
```

```
    {
        // Roman Urdu:
        // Har iteration mein marks add ho rahe hain

        sum = sum + marks[i];
    }

    // Roman Urdu:
    // Average return ki ja rahi hai

    return (float)sum / n;
}

// ========================================================
// Function: findMaximum()
// ========================================================

int findMaximum(int marks[], int n)
{
    // Roman Urdu:
    // Pehle element ko maximum maan liya

    int max = marks[0];

    // Roman Urdu:
    // Har value compare ho rahi hai

    for(int i = 1; i < n; i++)
    {
        // Roman Urdu:
        // Agar current value bari ho to max update hoga

        if(marks[i] > max)
        {
            max = marks[i];
        }
    }

    return max;
}

// ========================================================
// Function: findMinimum()
// ========================================================

int findMinimum(int marks[], int n)
{
    // Roman Urdu:
    // Pehle element ko minimum maan liya
```

```
    int min = marks[0];

    // Roman Urdu:
    // Har value compare ho rahi hai

    for(int i = 1; i < n; i++)
    {
        // Roman Urdu:
        // Agar current value choti ho to min update hoga

        if(marks[i] < min)
        {
            min = marks[i];
        }
    }

    return min;
}

// ====================================================
// Function: assignGrade()
// ====================================================

char assignGrade(int marks, float average)
{
    // Roman Urdu:
    // Relative grading average ke according apply ho rahi hai

    if(marks >= average + 15)
    {
        return 'A';
    }
    else if(marks >= average + 5)
    {
        return 'B';
    }
    else if(marks >= average - 5)
    {
        return 'C';
    }
    else if(marks >= average - 15)
    {
        return 'D';
    }
    else
    {
        return 'F';
    }
}

// ====================================================
```

```
// Function: displayResult()
// ========================================================

void displayResult(int marks[], int n, float average)
{
    cout << "\n========== FINAL RESULT ==========\n";

    // Roman Urdu:
    // Ye loop har student ka marks aur grade display karega

    for(int i = 0; i < n; i++)
    {
        // Roman Urdu:
        // assignGrade function grade return karega

        char grade = assignGrade(marks[i], average);

        cout << "Student " << i + 1
             << " | Marks = " << marks[i]
             << " | Grade = " << grade
             << endl;
    }
}
```

Sample Output

```
Enter number of students: 5

Enter marks of student 1: 85
Enter marks of student 2: 70
Enter marks of student 3: 60
Enter marks of student 4: 45
Enter marks of student 5: 30

========== FINAL RESULT ==========

Student 1 | Marks = 85 | Grade = A
Student 2 | Marks = 70 | Grade = B
Student 3 | Marks = 60 | Grade = C
Student 4 | Marks = 45 | Grade = D
Student 5 | Marks = 30 | Grade = F

============================
Class Average = 58
Maximum Marks = 85
Minimum Marks = 30
============================
```